

Nonlinear Dynamics in Computer Graphics: An Exploration of Reaction Diffusion Systems in Procedural Generation

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4th September 2026

Abstract

My passion and exposure to video games serves as the motivation for why I chose this as the focus of my research project. Through this research project, I'm aiming to investigate the role of Nonlinear Dynamics and Chaos in *Computer Graphics*. More particularly, I will be studying the mathematical construction of *Procedural Generation algorithms* that are used in generating *lifelike* landscapes and vegetation in 3D and creating texture patterns. Although the standard method for procedural generation relies on noise functions [2], I will be exploring an alternative method to procedural generation using *Nonlinear Reaction-Diffusion Systems* which is a mathematical model typically applied in biology and chemistry [1].

The above description is only indicative and I'm still exploring the viability of this topic for my research project. In the coming weeks, I will be doing more research on this topic and hoping to draft a thorough description of what I'm aiming to achieve in this research project. Any feedback or advice from teaching staff on whether this topic fits the content covered in this course would be greatly appreciated.

References

- [1] Roman Cherniha and Vasyl'Davydovych. *Nonlinear reaction-diffusion systems*. Springer, 2017.
- [2] A. Lagae et al. "A Survey of Procedural Noise Functions". In: *Computer Graphics Forum* 29.8 (2010), pp. 2579–2600. DOI: <https://doi.org/10.1111/j.1467-8659.2010.01827.x>. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1467-8659.2010.01827.x>. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1467-8659.2010.01827.x>.